

2(a)(i)	<i>any 3 from:</i> 1 percentage of Bt cotton has increased (by 72.9 / 73%) ; 2 (since 2007) the percentage of 1 gene Bt has decreased and 2 or more gene increased ; 3 number of hectares of cotton grown has increased (by 0.7 million hectares) ; 4 the use of pesticide has decreased as Bt cotton has increased ; 5 AVP ;	3
2(a)(ii)	<i>idea of</i> (less insecticide used) as more of the crop is protected from bollworm by the toxic proteins / Cry1Ac / Cry2Ab / AW ;	1
2(b)(i)	<i>idea of</i> the yield will decrease as the population of resistant bollworms will increase ; A in terms of greater selection pressure so only resistant insects survive	1
2(b)(ii)	<i>any 2 from:</i> 1 <i>idea of</i> non-Bt cotton enables non-resistant bollworms to survive ; 2 <i>idea of</i> breeding between susceptible and resistant bollworms keeps frequency of resistance genes lower (in the gene pool) ; 3 A in terms of less selection pressure so more susceptible survive to breed AVP ;	2
2(c)(i)	<i>independent variable:</i> the type of cotton plant / transgenic <i>vip3AcAa</i> / whether the cotton produces <i>Vip3AcAa</i> ; <i>dependent variable:</i> the number of dead larvae ;	2
2(c)(ii)	<i>any 1 from:</i> 1 number of larvae on each leaf ; 2 age of larvae ; 3 time (after adding larvae / for feeding) before counting dead larvae; 4 AVP ;	1

Question	Answer	Marks
2(c)(iii)	<i>any 2 from:</i> <i>idea of</i> control using non-Bt plant gives a baseline of larval death caused by factors other than toxic proteins in leaves ; <i>idea of</i> control using Bt <i>cry1Ac</i> plants for comparison with non-Bt to show, the larvae are Cry1Ac resistant / no difference in the number that die ; - AVP ;	2
2(c)(iv)	- difference between the number killed with Bt <i>cry1Ac</i> + vip3AcAa cotton (containing the toxic protein Vip3AcAa) is significant for both types of larvae <u>and</u> (as) standard deviations do not overlap with Bt <i>cry1Ac</i> cotton (which does not contain the toxic protein Vip3AcAa) ; - the difference between the increase in deaths of type A larvae and type B larvae killed on Bt <i>cry1Ac</i> + vip3AcAa (compared to those killed on Bt <i>cry1Ac</i>) is not significant <u>and</u> (as) standard deviations overlap ;	2
2(d)	<i>any 1 from:</i> <i>idea of</i> carrying out same investigation using susceptible strains of bollworm (as well as resistant types) ; - AVP ;	1